From the canonical source cone to the full quantum polar

Eight CHSH orientations recover every valid Bell inequality from one canonical source cone. This reduction is exact in both directions: combining orientations cannot repair a missing canonical certificate. We establish this geometry first and then convert a canonical semidefinite representation into a lift of the full quantum set.

The expectation vector and its homogenization are

$$q = (A_0, A_1, B_0, B_1, E_{00}, E_{01}, E_{10}, E_{11}), \qquad z = (1, q).$$

Let $P = \{f \in \mathbb{R}^9 : f^T z \geq 0 \text{ for every } q \in \mathcal{Q}_{222}\}$.

The canonical map. Use $H = 1 - (E_{00} - E_{01} + E_{10} + E_{11})/2$ and

$$4p(a, b \mid x, y) = 1 + aA_x + bB_y + abE_{xy}.$$

In the source order $c = (d, \lambda, n, m, C, A, D, \alpha, \beta)$, the event labels are

	λ	n	m	C	A	D	α	β
p_i	$++\|01$	$+-\|00$	$+-\|11$	$-+\|10$	$-+\|00$	$-+\|11$	$--\|01$	$+-\|10$

and $f_c = dH + 4\sum_i c_i p_i$. In the expectation order above, with the constant first, the nine coefficients $f = Mc$ are exactly

$$\begin{aligned}
f_0 &= d + \lambda + n + m + C + A + D + \alpha + \beta, \\
f_{A_0} &= \lambda + n - A - \alpha, & f_{A_1} &= m - C - D + \beta, \\
f_{B_0} &= -n + C + A - \beta, & f_{B_1} &= \lambda - m + D - \alpha, \\
f_{E_{00}} &= -d/2 - n - A, & f_{E_{01}} &= d/2 + \lambda + \alpha, \\
f_{E_{10}} &= -d/2 - C - \beta, & f_{E_{11}} &= -d/2 - m - D.
\end{aligned} \tag{1}$$

The map is invertible, with $\det M = -32$. Its inverse has a direct evaluation formula. Let $r = (0, 0, 0, 0, 1, -1, 1, 1)$ be the canonical PR point and $h = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$. For the losing event $(a, b \mid x, y)$, its deterministic neighbour is specified explicitly by

$$A_x = a, \quad B_y = b, \quad B_{1-y} = h_{x,1-y}a, \quad A_{1-x} = h_{1-x,y}b, \qquad E_{ij} = A_i B_j.$$

These neighbours obey $H(v_i) = 0$ and $p_j(v_i) = \delta_{ij}$; every event p_i vanishes at r. Hence

$$d = -f(r), \qquad c_i = f(v_i)/4. \tag{2}$$

Equivalently, if V has rows $(1, r)$ and $(1, v_i)$, then $VM = \mathrm{diag}(-1, 4I_8)$ and $M^{-1} = \mathrm{diag}(-1, \frac{1}{4}I_8)V$.

Eight exact relabelings. For $(r_0, r_1, s_0) \in \{\pm 1\}^3$, flip the outputs by $A_0 \mapsto r_0 A_0$, $A_1 \mapsto r_1 A_1$, $B_0 \mapsto s_0 B_0$, leaving B_1 unchanged. Both homogeneous expectations and coefficient vectors transform by

$$R_g = \mathrm{diag}(1, r_0, r_1, s_0, 1, r_0 s_0, r_0, r_1 s_0, r_1). \tag{3}$$

These are involutions preserving $\mathcal{Q}_{222}$. They send the canonical PR correlation pattern to

$$(r_0 s_0, -r_0, r_1 s_0, r_1),$$

giving all eight patterns with product -1. Thus the complete rational maps for chart g are

$$M_g = R_g M, \qquad c = M^{-1} R_g f, \qquad M_g^T z = M^T R_g z. \tag{4}$$

No input interchange or party swap is required.

Proposition 1. *Let* $C_{\mathrm{can}} = \{c \geq 0 : Mc \in P\}$. *Then*

$$P = \sum_{g \in \{\pm 1\}^3} M_g C_{\mathrm{can}}, \tag{5}$$

where the right side is a Minkowski sum of eight convex cones. Every $f \in P$ *belongs to one summand if it is negative at a PR point; otherwise it is a sum of elements from two opposite summands.*

Proof. The nonsignalling polytope has the sixteen deterministic vertices and the eight PR vertices. One short proof uses its sixteen probability inequalities. If all marginals are interior, each 2×2 outcome table has at most two zeros. A vertex needs eight independent active constraints, so every table has two opposite zeros and gives a perfect correlation or anticorrelation. An odd cycle parity forces all marginals to vanish, giving a PR point; even parity leaves a nontrivial segment of local mixtures. If any marginal is deterministic, sampling the other party's outcomes conditionally on the remaining local variable gives a local model, so such a vertex is deterministic.

Two distinct PR points have a local midpoint. If their patterns are opposite, this is the uniform local correlation. Otherwise they agree on two edges of the measurement square; the uniform distribution over the four deterministic assignments satisfying those two edge relations has the required midpoint. Therefore a locally nonnegative f can be negative at at most one PR point.

If $f \in P$ is negative at $R_g r$, then $R_g f$ is valid and negative at r. Formula (2) gives $d > 0$ and $c_i \geq 0$, because every deterministic neighbour lies in $\mathcal{Q}_{222}$. Thus $f = M_g c$ with $c \in C_{\mathrm{can}}$. If f is nonnegative at every PR point, it is nonnegative on the whole nonsignalling polytope. Polyhedral cone duality writes it as a nonnegative combination of the sixteen event probabilities. The losing events at r and at $-r$ partition those sixteen events. Each event generator is therefore the image of a source with $d = 0$ in one of those two charts. Such sources are in C_{can}. This proves the reverse inclusion in (5); the forward inclusion follows from quantum validity and physical relabeling. □

An individual canonical cone does not contain all valid slacks: every $M_g c$ with $c \geq 0$ takes value $-d \leq 0$ at its own PR point, whereas the constant slack 1 is positive at all eight. The sum in (5) is essential.

Theorem 2 (Exactness of the eight-orientation assembly)**.** *Let* $K \subseteq C_{\mathrm{can}}$ *be a closed convex cone containing every nonnegative source with* $d = 0$. *Put* $P_K = \sum_g M_g K$ *and* $Q_K = \{q : (1, q) \in P_K^*\}$. *Then* P_K *is closed and*

$$P_K \cap M(\mathbb{R}^9_+) = MK.$$

Moreover

$$\mathcal{Q}_{222} \subseteq Q_K \subseteq \mathrm{NS}_{222}, \qquad Q_K = \mathcal{Q}_{222} \iff K = C_{\mathrm{can}}.$$

Proof. The set inclusions follow from $P_K \subseteq P$ and the presence of all sixteen event generators in P_K. If $Mc \in P_K$ with $c \geq 0$ and $d = 0$, then $c \in K$ by assumption. For $d > 0$, write $Mc = Mc^0 + \sum_{g \neq 0} f_g$, where $c^0 \in K$ and $f_g = M_g c^g$. At the own PR point r_g, $f_g(r_g) = -d_g \leq 0$. Local midpoint positivity gives $f_g(r) \geq d_g \geq 0$. In the base source coordinates, each f_g therefore has d-coordinate at most zero and event coordinates $f_g(v_i)/4 \geq 0$. Write $c = (d, a)$ and $c^0 = (d_0, a_0)$. Then $0 < d \leq d_0$ and $a \geq a_0$. Thus, with $\theta = d/d_0$,

$$c = \theta c^0 + (0, a - \theta a_0) \in K.$$

The reverse implication is immediate. For a convergent sequence of sums in P_K, their constant coefficients are $\sum_g \sum_k c_k^g$. Nonnegativity bounds each source sequence; closedness of K gives convergent subsequences and a sum in P_K. If $K = C_{\text{can}}$, Proposition 1 gives $Q_K = \mathcal{Q}_{222}$. Otherwise every missing valid source has $d > 0$ and so gives a valid slack outside the closed cone P_K. A strict separating vector z is in P_K^*. Since P_K contains all probability generators, z is in the homogenized nonsignalling cone; it is nonzero and has $z_0 > 0$. Normalizing $z_0 = 1$ gives a point outside $\mathcal{Q}_{222}$. □

Exact dualization. Let a homogeneous canonical SDP cone be given by

$$\mathcal{L}(c, y) = \sum_{k=0}^{8} c_k A_k + \sum_{j=1}^{p} y_j B_j, \qquad K = \{c : \exists y,\ \mathcal{L}(c, y) \succeq 0\},$$

where $y \in \mathbb{R}^p$. Suppose its total block order is n and $\mathcal{L}(c_*, y_*) \succeq \epsilon I_n$ for some $\epsilon > 0$. Then

$$K^* = \{\mathcal{A}^* Y :\ Y \succeq 0,\ \mathcal{B}^* Y = 0\}, \qquad (\mathcal{A}^* Y)_k = \operatorname{Tr}(A_k Y), \quad (\mathcal{B}^* Y)_j = \operatorname{Tr}(B_j Y), \tag{6}$$

with Y block diagonal in the same orders as $\mathcal{L}$ and no closure required. Indeed, a convergent sequence of full adjoint images $(\mathcal{A}^* Y_i, \mathcal{B}^* Y_i)$ satisfies

$$\epsilon \operatorname{Tr} Y_i \leq \langle Y_i, \mathcal{L}(c_*, y_*) \rangle = c_* \cdot \mathcal{A}^* Y_i + y_* \cdot \mathcal{B}^* Y_i.$$

This bounds the positive semidefinite Y_i and gives a convergent subsequence. Thus the full adjoint image is closed. Taking the dual of the preimage cone and then the section $\mathcal{B}^* Y = 0$ proves (6).

Theorem 3 (Assembly from an exact canonical cone)**.** *Suppose $K = C_{\text{can}}$ and the homogeneous pencil for K has a strict feasible point. Then the full quantum set is exactly*

$$\mathcal{Q}_{222} = \{q :\ \exists Y_g \succeq 0 \text{ for every } g, \quad \mathcal{B}^* Y_g = 0, \quad \mathcal{A}^* Y_g = M_g^T (1, q)\}. \tag{7}$$

It uses eight copies of the canonical dual blocks, of total order $8n$. If the canonical pencil is rational, so is this representation.

Proof. By (6), the constraint for chart g is exactly $M_g^T z \in C_{\text{can}}^*$. All eight constraints are therefore equivalent to $z \in (\sum_g M_g C_{\text{can}})^* = P^*$ by Proposition 1. Since $\mathcal{Q}_{222}$ is compact and convex, $P^* \cap \{z_0 = 1\} = \{(1, q) : q \in \mathcal{Q}_{222}\}$. The eight Gram copies and their linear adjoint equations give the claimed representation. These equations can be eliminated by fixed linear algebra. There are no residual affine restrictions on q: the homogenizations of the full-dimensional quantum set span $\mathbb{R}^9$, so the prescribed adjoint targets are linearly consistent for every q. □

For the quadratic half-line certificate cone, the exact adjoint equations give an outer approximation with total order 264 and 384 auxiliary variables. Its strict feasible point, rational coordinate maps and eliminated equations are supplied in the accompanying data. Theorem 2 identifies its precise scope: it equals the quantum set exactly when the canonical half-line cone contains every valid source.